Prioritized memory transfer: a predictive-coding primitive for continual learning

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Abstract

Biological memory is consolidated during sleep, when the brain replays stored experience with no external teacher and, remarkably, appears to favour what has not yet been consolidated. Reproducing this in artificial networks is the crux of continual learning: how can a network rehearse its own memories, and how can that rehearsal be *prioritized* toward what a downstream network still lacks, without an external scheduler holding the list of what to replay? We show that predictive coding answers both questions at once. Two linear associative memories are coupled in a hierarchy: a frozen *teacher* above an empty, plastic *student*. Flipping the sign of a single interface precision turns ordinary recall into replay. Under this "reversed precision" the student's own prediction error drives the teacher into exactly the memories the student cannot yet predict; the student learns them; the drive there vanishes; the process stops itself. We prove that the drive is the gradient of the student's own surprise, so priority is not imposed but derived — the student's error *is* the priority signal — and that at a precise matching condition the coupled dynamics form an exact free-energy saddle. Because the substrate is linear, transfer acquires a memory *subspace* rather than named patterns. Transfer is a primitive that composes: interleaving two teachers merges their subspaces and cancels the crosstalk between correlated memories, and a buffer $\rightarrow$ consolidate $\rightarrow$ storage loop built on it retains a streamed sequence of correlated memories where a buffer-only control catastrophically forgets. The result is a compact, analytically transparent, biologically suggestive account of prioritized replay and its use for continual learning.

Author summary

When we sleep, the brain spontaneously replays recent experience, and this replay helps move memories into long-term storage without erasing what is already there. Artificial neural networks are notoriously bad at this: teaching them something new tends to overwrite the old, a failure called catastrophic forgetting. Rehearsing old memories fixes it, but only if the network can somehow retrieve those memories by itself and, ideally, spend its effort on the memories that still need consolidating. We describe a simple two-network model in which this happens automatically. A frozen "teacher" network holds memories; an empty "student" network learns them. By reversing the sign of a single connection parameter, the pair switches from ordinary recall into a replay mode in which the teacher is pushed toward whatever the student has not yet learned — and

the push is generated entirely by the student's own prediction error, so no outside controller is needed to decide what to replay. We prove that this push always climbs the student's "surprise" and switches itself off once a memory is learned. Stacking this transfer step into a small loop lets a network absorb a stream of new, overlapping memories while keeping the earlier ones, offering a biologically plausible route around catastrophic forgetting.

1 Introduction

A system learns continually when it can absorb new information over time without discarding what it already knows. Artificial neural networks fail at this in a characteristic way: training on a new task or pattern in isolation rapidly erodes the representations built for earlier ones, the phenomenon of *catastrophic forgetting* [1–3]. In associative memories the failure is concrete — storing a new activity pattern sequentially degrades the previously stored ones.

The oldest and most robust remedy is *rehearsal*: periodically retrain the network on its earlier memories so their representations are reinforced while new ones are added [2]. Rehearsal has a striking biological counterpart. During sleep and quiet waking the hippocampus spontaneously reactivates recent experience, and this replay is causally implicated in memory consolidation [4, 5]. The complementary-learning-systems account casts a fast hippocampal store and a slow neocortical store in exactly the teacher/apprentice roles that rehearsal requires: the fast system replays to train the slow one [6, 7]. Two questions, however, are usually left open. First, rehearsal presupposes that the network can *retrieve* its own memories to replay them, rather than reading them from an external list. Second, replay is not uniform — consolidation should concentrate on what the slow system has not yet acquired. *How is autonomous replay generated, and how is it prioritized, without an external scheduler?*

In earlier work we addressed the first question within a single continuous Hopfield network: plastic self-inhibition lets the network shallow out already-visited attractors and so recover its full set of stored patterns from its own dynamics, which is enough to drive a continual-learning loop [8]. Here we address the second, and we do so by moving from one network to two coupled by *predictive coding*. Predictive coding models a cortical hierarchy as levels that each try to predict the level below and pass up only the residual error, learning by descending a variational free energy through purely local computations [9–11]; Tang, Barron and Bogacz recently showed that a recurrent predictive-coding network with covariance learning is a faithful, biologically grounded associative memory [12], and that substrate is the one we build on. Coupling two such memories exposes a quantity a single network does not have: the prediction error of one network about the other. That error is precisely a measure of what one network cannot yet predict, and we use it as the priority signal — an idea that mirrors prioritized experience replay in reinforcement learning, where transitions are replayed in proportion to their error [13], but which here falls out of the network's own free energy rather than being engineered. The free-energy reading also connects the mechanism to active inference, in which agents act to minimize (variational) free energy [14, 15]; we will draw that connection precisely where it is exact and flag it as merely structural where it is not [16].

Our contribution is one mechanism presented as a ladder of three rungs, each built on the one below.

1. **Transfer.** One teacher T drives one student S. A single signed interface precision selects the phase: positive is wake/recall, negative is sleep/replay. In replay the student's prediction error steers the teacher into the memories the student lacks; the student learns them; the drive self-extinguishes. We prove the drive is gradient ascent on the student's own surprise, and that the coupled flow is an exact free-energy saddle at a precise precision-matching condition.

2. **Union of subspaces.** Because transfer is a reusable primitive, coupling a student to two teachers in alternation merges their memory subspaces. Interleaving one teacher per replay bout cancels the crosstalk that correlation would otherwise create between the two sets of memories.

3. **Continual learning.** A small buffer → consolidate → storage loop, with the consolidation step performed by interleaved transfer, retains a stream of correlated memories where a buffer-only control forgets all but the most recent.

Throughout, the substrate is a *linear* covariance predictive-coding memory. Linearity is a deliberate simplification: it makes the whole mechanism analytically transparent, at the cost that what transfers is a memory *subspace* (an effective rank) rather than a set of named episodes. We state this scope plainly and return to it in the Discussion.

2 Model and analysis

We first fix the model and then state its formal properties. Following Tang et al. [12] and the predictive-coding convention, results are stated here and proved in full in Materials and methods.

2.1 A two-population predictive-coding memory

Both networks live in the same space $\mathbb{R}^d$, one coordinate per neuron. Each is a linear predictive-coding associative memory: a population with recurrent weights W (zero diagonal, so no neuron predicts itself) tries to reconstruct its own state, producing the *mismatch*

$$M = I - W, \qquad \varepsilon = Mx, \qquad S = M^\top M, \tag{1}$$

where ε is the reconstruction error and S, its Gram matrix, is the symmetric positive semi-definite *self-error operator*. A pattern m is stored when it is a fixed point of the reconstruction, $Mm = 0$; the stored memories therefore span the kernel of M, a flat linear manifold (house identity H1–H2 of the accompanying notes).

We couple a **teacher** T and a **student** S (Fig 1). The teacher sits higher in the hierarchy: it is a fully trained memory, its weights W_T frozen, and it is confined to its own memory manifold. The student sits below: it starts empty ($W_S = 0$) and is plastic. Three errors drive the pair — the teacher's self-error $\varepsilon_T = M_T x_T$, the student's self-error $\varepsilon_S = M_S x_S$, and the *interface error*

$$\varepsilon_{TS} = x_S - x_T, \tag{2}$$

the disagreement between the two states, which the identity interface reports at the student's level. Each error is weighted by a precision (an inverse variance): $\pi_T, \pi_S > 0$ are the self precisions, $\pi_{TS} > 0$ is the precision with which the student trusts its input from the teacher, and π_{ST} is the teacher's interface precision — the one quantity we allow to be *signed*. The state and weight dynamics are

$$\tau_T\, \dot{x_T} = -\,\pi_T M_T^\top \varepsilon_T \;+\; \pi_{ST}\, \varepsilon_{TS} \;+\; \xi, \tag{3}$$

$$\tau_S\, \dot{x_S} = -\,\pi_{TS}\, \varepsilon_{TS} \;-\; \pi_S M_S^\top \varepsilon_S, \tag{4}$$

$$\dot{W}_S = \eta\, \pi_S\, \varepsilon_S\, x_S^\top, \qquad \operatorname{diag}(W_S) \equiv 0, \tag{5}$$

with ξ a small isotropic noise on the teacher and the timescales ordered $\tau_S \ll \tau_T \ll 1/\eta$: the student perceives fast, the teacher drifts slowly, the weights change slowest of all. The teacher is additionally held to a fixed norm, so it can reorient but not run away.

Reversed precision: one knob for wake and sleep. Everything hinges on the sign of π_{ST} in Eq. (3). When $\pi_{ST} > 0$ the teacher, like the student, is pulled to *reduce* the interface error: both networks seek agreement, the student completes the teacher's pattern, and nothing is transferred. This is ordinary recall — *wake*. When $\pi_{ST} < 0$ the

[figure placeholder]

Two coupled predictive-coding memories. *Frozen teacher T (above) over an empty, plastic student S (below); each has recurrent weights W and a self-error population (ε_T, ε_S), and the identity interface reports the disagreement $\varepsilon_{TS} = x_S - x_T$. The teacher's interface precision π_{ST} is signed: $\pi_{ST} > 0$ (wake/recall) pulls both toward agreement — no transfer; $\pi_{ST} < 0$ (sleep/replay) reverses the teacher's drive, pushing it toward whatever the student cannot predict. One sign is the whole phase.*

Fig 1. Model architecture and the wake/sleep sign flip. See Section 2.1.

same term becomes a *reversed precision*: the teacher is pushed to *increase* the disagreement, a drive to seek out states the student cannot match. This is *sleep*, or replay. No separate gate switches the phase; the sign of a single precision *is* the phase. This is the paper's pivot, and the rest of the analysis simply works out its consequences.

Surprise and free energy. Each network carries a free energy, the precision-weighted sum of its squared errors,

$$F_S = \tfrac{\pi_{TS}}{2}\,\|\varepsilon_{TS}\|^2 + \tfrac{\pi_S}{2}\,\|\varepsilon_S\|^2, \qquad F_T = \tfrac{\pi_T}{2}\,\|\varepsilon_T\|^2, \tag{6}$$

the network's *surprise* at its current state. We keep two words strictly apart throughout, and the reader will save themselves confusion by doing the same:

Remark 1 (Novelty versus surprise). *Novelty* is a property of the student's *weights*: which directions it cannot yet predict. It is carried by an operator, N_S below, and its eigenvalues n_k. *Surprise* is a property of a *state*: how badly the current configuration is predicted, measured by the scalar free energies F_S, F_T. Novelty is the spectrum; surprise is the score. There is no "teacher's surprise" other than F_T.

Two inequalities delimit the regime we operate in, and we state them as such rather than as hypotheses of the theorems: the input guard $\pi_{TS} > \pi_S$ (the student trusts its teacher input over its own recurrent prior, which prevents it from confabulating), and a stability guard on the *magnitude* of the reversed precision, made precise in Lemma 2.

2.2 Inference and learning descend a free energy

The first property justifies calling F_S a free energy at all: the student's dynamics are gradient descent on it.

Proposition 1 (Free-energy descent). *The student's state relaxation* (4) *is exact gradient descent on F_S, $\tau_S \dot{x_S} = -\nabla_{x_S} F_S$, and its zero-diagonal weight update* (5) *is projected gradient descent on F_S; along both flows F_S is non-increasing.*

The state gradient is $\nabla_{x_S} F_S = \pi_{TS}\varepsilon_{TS} + \pi_S M_S^{\top}\varepsilon_S$, which is exactly minus the right-hand side of Eq. (4); the weight gradient is $\nabla_{W_S} F_S = -\pi_S \varepsilon_S x_S^{\top}$, and removing its diagonal is an orthogonal projection, so descent survives it (H9; proof in Materials and methods, verified numerically to machine precision).

2.3 The novelty operator

Because the student is fast ($\tau_S \ll \tau_T$), on the teacher's timescale it always sits at its own equilibrium. Solving $\nabla_{x_S} F_S = 0$ for the settled student gives the operator that governs everything downstream.

Lemma 1 (Novelty operator)**.** *At the fast-student equilibrium the student's state is* $x_S^* = (I - N_S)\, x_T$ *and the interface error is* $\varepsilon_{TS} = -N_S x_T$*, with*

$$N_S \;=\; \pi_S S_S \,(\pi_{TS} I + \pi_S S_S)^{-1}. \tag{7}$$

N_S *is symmetric positive semi-definite, shares the eigenbasis of* S_S*, and annihilates every learned direction: if* $M_S u = 0$ *then* $N_S u = 0$*.*

The reading is clean. The student copies back the part of the teacher's state it can predict, $x_S^* = (I - N_S)x_T$, and leaves the part it cannot, so the residual disagreement $\varepsilon_{TS} = -N_S x_T$ is *exactly the novel part of the teacher's state* (H5–H6). In the shared eigenbasis $N_S u_k = n_k u_k$ with novelty eigenvalues

$$n(\mu) = \frac{\pi_S\, \mu}{\pi_{TS} + \pi_S\, \mu} \in [0, 1), \tag{8}$$

where $\mu \geq 0$ is the corresponding eigenvalue of S_S: a fully learned direction ($\mu = 0$) has $n = 0$, and novelty grows monotonically with how badly the student mispredicts the direction. N_S is a *novelty meter* built from the student's weights alone.

2.4 The surprise identity

Novelty (a weight-level spectrum) and surprise (a state-level scalar) are not two mechanisms but two readings of one quantity. This is the bridge, and it is exact.

Corollary 1 (Surprise identity)**.** *At the fast-student equilibrium the student's surprise about the teacher's state is the teacher's state scored by the novelty meter:*

$$F_S^{\mathrm{eq}}(x_T) \;\equiv\; \min_{x_S} F_S(x_S, x_T) \;=\; \tfrac{\pi_{TS}}{2}\, x_T^\top N_S\, x_T, \qquad \textit{hence} \qquad \nabla_{x_T} F_S^{\mathrm{eq}} = \pi_{TS}\, N_S\, x_T. \tag{9}$$

The proof is a two-line computation — per eigendirection the interface term contributes n^2 and the self term $n(1-n)$, which sum to n (H8) — or a one-line application of the envelope theorem, since x_S^* minimizes F_S. Either way, the settled student's free energy is literally $\sum_k n_k c_k^2$ with $c_k = u_k^\top x_T$: novelty priced out on where the teacher's state actually sits. The gradient we will need in Theorem 1 follows immediately.

2.5 The student steers the teacher up the student's surprise

We can now read the teacher's sleep dynamics. Substituting the settled interface error $\varepsilon_{TS} = -N_S x_T$ into Eq. (3) and writing $\pi_{ST} = -|\pi_{ST}|$ for the reversed precision,

$$\tau_T\, \dot{x_T} \;=\; \underbrace{-\,\pi_T S_T\, x_T}_{\text{holds its own memory}} \;+\; \underbrace{|\pi_{ST}|\, N_S\, x_T}_{\text{the student's push } u}\;, \tag{10}$$

the teacher's own self-pull plus a push $u = |\pi_{ST}| N_S x_T$ produced entirely by the student. This push is the heart of the paper.

Theorem 1 (Prioritized transfer)**.** *The student's push* $u = |\pi_{ST}|\, N_S\, x_T$ *has three equivalent readings.* (i) Per direction: *in the student's eigenbasis each component of the teacher's state grows at a rate set by that direction's novelty,* $\dot{c}_k = (|\pi_{ST}|/\tau_T)\, n_k\, c_k$*; a learned direction (*$n_k = 0$*) is frozen, a novel one is amplified, and the more novel, the faster.* (ii) Steepest ascent: *the push points along the gradient of the student's surprise,* $u = (|\pi_{ST}|/\pi_{TS})\, \nabla F_S^{\mathrm{eq}}$*, hence it is the direction of fastest increase of* F_S^{eq}*.* (iii) Self-limiting: *as the student learns a direction (*$\|M_S u_k\| \to 0$*, by Proposition 1) its* *novelty $n_k \to 0$, so the push there fades to zero. Consequently the full teacher flow is $\tau_T \dot{x_T} = -\nabla F_T + (|\pi_{ST}|/\pi_{TS})\, \nabla F_S^{\mathrm{eq}}$: the teacher descends its own surprise while ascending the student's.*

In words: the teacher's state is passed through the student's novelty meter, and the meter amplifies whatever the student finds novel and ignores whatever it has learned. Reading (ii) says this is not an arbitrary drift but the steepest local climb of the student's own surprise — priority is *derived*, not imposed. Reading (iii) says the climb flattens the very landscape it climbs: every direction the teacher is dragged into gets learned, its hill sinking to a plain, so the process stops itself when the student has acquired everything the teacher holds. Two honest caveats travel with the theorem: the ascent is local (steepest from where the teacher currently is, not a beeline to the globally most novel direction), and a direction the teacher does not currently occupy ($c_k = 0$) receives no push — it is the noise ξ that seeds those, and the reversed-precision drive that then amplifies them.

2.6 Staying on the memory manifold

The push must not tip over into chasing noise off the manifold. The teacher's self-pull prevents this, provided the reversed precision is not too strong.

Lemma 2 (Off-manifold stability)**.** *Under the conservative guard $|\pi_{ST}| < \pi_T\, \sigma^2_{\min}$, where $\sigma^2_{\min}$ is the smallest nonzero eigenvalue of S_T, the block of the dynamics normal to the teacher's memory manifold is strictly contracting, so the teacher consolidates on-manifold and stays quiescent. Above the guard it is driven off-manifold into noise. (The guard is not exact manifold invariance; residual cross-block leakage is small and measured numerically.)*

This is what sets the precision-weighting regime: the reversed precision must be strong enough to move the teacher along novel memory directions, yet weak enough that the self-pull wins normal to the manifold.

2.7 An exact free-energy saddle

Both forces in Theorem 1 are gradients, so it is natural to ask when the coupled (x_T, x_S) flow is generated by a *single* potential. It is, at one precise condition.

Proposition 2 (Free-energy saddle)**.** *The sleep dynamics are generated by the single functional $\Phi = F_T - F_S$ — the teacher descending Φ, the student ascending it — if and only if*

$$\pi_{ST} = -\pi_{TS}. \tag{11}$$

Equivalently (Theorem 1(ii)), the teacher ascends the student's surprise at exactly the rate π_{TS} that defines it: the ascent's coefficient $|\pi_{ST}|/\pi_{TS}$ equals one. At this point the pair plays a zero-sum game on one potential — the student descending F_S (Proposition 1), the teacher climbing it — a minimax whose shape is reminiscent of the agent–environment saddle of active inference, though we do not claim identity of the two [15, 16].

Remark 2 (Scope: subspace, not named episodes)*.* The substrate is linear, so a teacher's memories span a flat subspace and what transfers is that subspace — an effective rank, not a labelled list of patterns. Episodic, one-pattern-per-step replay would require an added nonlinearity (e.g. a soft winner-take-all); we return to this in the Discussion.

3 Results

We now show the mechanism at work and then that it composes. The first two subsections establish the single-teacher primitive (rung 1) and its free-energy reading; the last two climb the ladder to the union of subspaces (rung 2) and continual learning (rung 3). All simulations use the linear covariance predictive-coding substrate of Section 2.1; parameters and diagnostics are in Materials and methods.

3.1 Prioritized transfer of unlearned directions

We start with the picture. Fig 2 takes a single teacher holding two memories and looks at the teacher's state in the plane spanned by one direction the student has *already* learned and one it has *not*. The reversed-precision flow tells the whole story at a glance: along the learned axis the teacher is pinned — the push there is null and the self-pull holds it in place — while along the novel axis it is driven outward, and a sample trajectory (red) rides the novel direction until the student catches up. This is Theorem 1(i) made visible: motion only along what the student does not yet know. The figure is the two-network analogue of the attractor stream plots used for single associative memories [12] and in our own earlier network [8], and reading it the same way is intended.

[figure placeholder]
Fig 2 source: a short script on the two-population model — project $\tau_T \dot{x}_T$ from Eq. (10) onto the (learned, novel) plane; overlay one transfer trajectory. Analogous to the CHN vector-field figure.

[figure placeholder]
The reversed-precision flow pushes the teacher only along unlearned directions. *Teacher state in a 2D slice: horizontal axis a memory direction the student has learned, vertical axis one it has not. Arrows show $\tau_T \dot{x}_T$ in replay ($\pi_{ST} < 0$). Along the learned axis the flow is inward/pinned (push $= 0$, self-pull holds); along the novel axis it is outward (push $= |\pi_{ST}|\, n\, c$). Red: one transfer trajectory, riding the novel axis until learning flattens it.*

Fig 2. Prioritized replay is visible as a flow field. See body text, Section 3.1.

The quantitative claim is that transfer acquires the teacher's whole memory subspace, prioritizing the directions the student lacks, and then stops. Fig 3 shows it. The novelty spectrum — the eigenvalues of N_S restricted to the teacher's memory manifold, $U_T^\top N_S U_T$ — starts high on every stored direction and falls to zero one direction at a time as the student learns them, producing a descending "staircase" (left). The accompanying alignment heatmap (right) shows the teacher's state progressively filling the memory manifold as the drive on each direction extinguishes. Crucially, the number of steps is the memory set's *effective rank*, not its pattern count: $P = 6$ orthonormal patterns give six steps, but P correlated patterns of rank three give only three, because learning one direction explains its correlated neighbours at once. This is the linear model transferring a *subspace* (Remark 2). We report reliable completion of the manifold across the regimes we tested; the exact step count and its ordering are a spectral, not a temporal, effect, and the apparent sharpness of the steps is partly a plotting artefact of the timescale.

[figure placeholder]
Left: *restricted novelty spectrum* $\mathrm{eig}(U_T^\top N_S U_T)$ *vs. replay time: a descending staircase, one step per learned direction; overlaid, orthonormal (*P=6 $\to$ 6 *steps) vs. correlated rank-3 (*$\to$ 3 *steps).* ***Right:*** *heatmap of* x_T*–manifold alignment over time, filling as the drive extinguishes. Source:* `notebooks/two_network/01_single_run.py`, `05_findingH_subspace.py`.

Fig 3. Transfer acquires the memory subspace and self-terminates; steps count effective rank. See Section 3.1.

3.2 The wake/sleep saddle picture

Why does the replay prioritize, and why does it stop? Section's theory answers both, and Fig 4 shows the answer directly. It plots cross-sections of the free-energy landscape the teacher moves on. Along a direction the student can already predict, the landscape is a *bowl*: the teacher sits at the bottom, held on the manifold. Along a direction the student cannot predict, it is a *hill*: the reversed precision turns that axis into an unstable ascent, and the teacher slides up it — into the novel memory. As the student learns the direction, the hill relaxes back into a bowl, and the drive is gone. Learning literally reshapes the landscape from hill to bowl, one direction at a time.

At the matching condition $\pi_{ST} = -\pi_{TS}$ (Proposition 2) these cross-sections are two faces of one potential $\Phi = F_T - F_S$: a genuine saddle, minimized by the teacher over learned directions and maximized over novel ones. We verify the condition quantitatively by measuring the circulation of the coupled flow, the obstruction to its being a single gradient; it vanishes precisely at $\pi_{ST} = -\pi_{TS}$ and grows linearly on either side, matching $|\pi_{ST} + \pi_{TS}|\sqrt{d}$ to machine precision.

Two further checks confirm the mechanism is the one we claim, and we relegate their figures to Supporting information. First, stability: sweeping the strength $|\pi_{ST}|$ shows the teacher consolidating on-manifold below the guard of Lemma 2 and abruptly chasing noise above it, with the turnover at the predicted value. Second, selection is spectral, not a race in time: across the entire fast-student regime a known direction is explained away while a novel one is not, and the discrimination collapses only as $\tau_S \to \tau_T$, i.e. only when the student is no longer fast — so it is what the student's weights can null, not raw speed, that decides.

3.3 Merging memories by interleaving

Transfer is a primitive, and primitives compose. If a student is coupled to two teachers, each holding its own memories, can it acquire the *union* of their subspaces? It can, provided the two teachers are replayed one at a time rather than together. We rehearse teacher T_1 for a bout, then T_2 for a bout, and alternate; each bout is a transfer step of the kind analysed above, with only one interface active. Over bouts the student accumulates $\mathcal{U}_1 + \mathcal{U}_2$.

The subtlety, and the reason for interleaving, is *crosstalk* between correlated memories. Fig 5 shows it as a sawtooth. Rehearsing T_1 drives the residual on T_1's

[figure placeholder]
Cross-sections of $\Phi = F_T - F_S$ at $\pi_{ST} = -\pi_{TS}$: a bowl along a predictable direction (teacher pinned) vs. a hill along a novel direction (teacher driven up it); a middle panel shows a hill relaxing to a bowl as the direction is learned. Source:
`notebooks/two_network/02_findingD_saddle.py`, `06_eigenspace_geometry.py`.

Fig 4. Replay is ascent on a hill that learning turns into a bowl. At $\pi_{ST} = -\pi_{TS}$ the two faces belong to one saddle potential $\Phi = F_T - F_S$. See Section 3.2.

memory to zero but, because the two memory sets overlap, nudges the residual on T_2's memory up; rehearsing T_2 then nulls its own and nudges T_1's back up. Left there, this is exactly the interference that makes sequential learning forget. But the bumps shrink each round — each rehearsal re-excites and re-cancels the correlated perturbation the other one caused — and the sawtooth decays to zero: both subspaces are held at once. This is interleaved rehearsal, the classical cure for catastrophic forgetting, arising here as the natural way to compose two transfer steps.

[figure placeholder]
Union deficit and per-teacher residuals vs. rehearsal bout: a decaying sawtooth — rehearsing T_1 nulls its memory and bumps T_2's, and vice versa, the bumps decaying to zero as both subspaces are jointly held. Source:
`notebooks/additive_synthesis/interleaved/01_interleaved_single_run.py`, `02_interleaved_edge_cases.py`.

Fig 5. Interleaved transfer merges two memory subspaces and cancels crosstalk. The sawtooth decays to zero: $\mathcal{U}_1 + \mathcal{U}_2$ is retained. See Section 3.3.

3.4 Continual learning without catastrophic forgetting

The top rung assembles transfer into a loop that absorbs a *stream* of memories (Algorithm 1). Three networks divide the labour: a fast **buffer** that takes in one new memory at a time (and is the only network that ever learns from external data), a transient **synthesis** workspace, and a slow **storage** that accumulates the consolidated result. Each cycle writes the new memory into the buffer, *consolidates* buffer-and-storage into synthesis by interleaved transfer (Section 3.3), then *downloads* synthesis into storage. The interleaving is what lets storage's existing memories be rehearsed alongside the new one, so old correlated memories are re-excited and preserved rather than overwritten.

Fig 6 shows the payoff. As memories stream in, we track a retention matrix: for each stored memory (row), how well it is still recovered after each cycle (column). Under the loop, the matrix fills and stays filled — every memory acquired so far remains retrievable, and the stored subspace grows one memory per cycle up to capacity. A buffer-only control, which simply writes each new memory without the consolidate/download loop, retains only the most recent: its retention matrix is a single

Algorithm 1 Continual learning by prioritized transfer

Input: stream of memories $m_1, m_2, \ldots$; empty storage W_S^{sto}
for each arriving memory m_k **do**
    Write: store m_k in the fast buffer (one-shot), overwriting the previous
    Consolidate: interleaved transfer of {buffer, storage} $\to$ synthesis (alternate rehearsing buffer and storage; Section 3.3)
    Download: transfer synthesis $\to$ storage
end for
Return: storage holding span$\{m_1, \ldots, m_k\}$ up to capacity $d - 1$

bright diagonal against a forgotten background — catastrophic forgetting in its purest form. The difference is the loop, and the loop is nothing but transfer, applied twice per cycle.

[figure placeholder]
Left: *retention matrix under Algorithm 1 (memory × cycle) — fills and stays filled; stored subspace grows one memory per cycle.* ***Right:*** *buffer-only control — only the latest memory retained (bright diagonal, forgotten elsewhere). Source:* `notebooks/continual_learning/01_continual_single_stream.py`.

Fig 6. The transfer loop retains a correlated stream where a buffer-only control forgets. See Section 3.4.

4 Discussion

We have shown that a single signed precision turns a pair of predictive-coding memories from a recall circuit into a replay circuit, and that in replay the downstream network's own prediction error steers the upstream network into exactly the memories still missing downstream. The priority is not scheduled; it is the gradient of the student's surprise (Theorem 1), and at a precise precision-matching condition the coupled dynamics are an exact free-energy saddle (Proposition 2). Transfer is a primitive that composes into a union of subspaces and, from there, into a continual-learning loop.

What is exact and what is only suggestive. The variational free-energy reading is exact where we use it: the student's inference and learning are gradient descent on F_S (Proposition 1), and the settled student's free energy is the novelty score the teacher climbs (Corollary 1). We are deliberately more cautious with two larger claims. First, that the loop minimizes a free energy *integrated over the whole data stream* is a conjecture, not a result — we prove per-step descent, not a global objective. Second, the resemblance of the $\pi_{ST} = -\pi_{TS}$ saddle to the agent–environment minimax of active inference is structural, a matter of shape; we do not claim the reversed precision implements an expected-free-energy computation [16]. Flagging these keeps the free-energy story honest.

Biological reading. The architecture maps naturally onto complementary learning systems: a fast teacher that already holds an experience (hippocampus) trains a slow, initially empty student (neocortex) offline, and does so preferentially for what the neocortex has not yet consolidated [4–7]. The wake/sleep distinction reduces to the sign of one interface precision, and prioritization requires no homunculus choosing what to replay — the error does the choosing, as in prioritized experience replay [13] but here derived from the network's own free energy. This complements our earlier single-network account, where autonomous replay was produced instead by plastic self-inhibition shallowing visited attractors [8]: there one network recovered its own memories; here two networks transfer them, and the priority signal is made explicit.

Limitations and outlook. The price of analytic transparency is linearity, and with it the scope of Remark 2: transfer moves a memory *subspace*, an effective rank, not a set of named episodes. Episodic, one-pattern-per-bout replay — the regime in which the biological analogy is sharpest — would need a nonlinearity such as a soft winner-take-all to select a single pattern per replay, and building that on the same reversed-precision skeleton is the natural next step. Manifold completion of the fully coupled dynamics is established empirically across the regimes we tested rather than proven, and a stochastic-approximation treatment of the noise-seeded exploration remains open. None of these affects the core claims, which are the exact ones above.

5 Conclusion

Prioritized memory transfer is a small, ownable primitive: reverse one interface precision, and a downstream network's prediction error drives an upstream network to replay exactly what the downstream network still lacks, stopping itself when the gap is closed. Because the mechanism is a gradient on the downstream network's own surprise, priority comes for free; because the substrate is a linear predictive-coding memory, the whole thing is exactly analyzable; and because transfer is a primitive, it composes — into a union of memory subspaces, and into a continual-learning loop that resists catastrophic forgetting. The account is compact enough to prove and biologically suggestive enough to motivate the nonlinear, episodic extensions it points to.

Materials and methods

Substrate and faithfulness. Each population is a linear covariance predictive-coding network with one tied weight matrix W (zero diagonal, enforced at construction and after every update): top-down prediction uses $M = I - W$, bottom-up error projection uses $M^\top$, and the teacher–student interface is identity in both directions. Memories are stored in the teacher as $M_T m_p = 0$ ($\max_p \|M_T m_p\| < 10^{-4}$ for $P \le d-1$). The student's perception equals $-\nabla_{x_S} F_S$ and its learning $-\nabla_{W_S} F_S$ (autograd-verified to $< 10^{-8}$).

Proofs. Proposition 1 (state and projected-weight gradient descent), Lemma 1 (fast-student elimination and N_S), Corollary 1 (the surprise identity, via per-direction summation $n^2 + n(1-n) = n$ and, independently, the envelope theorem), and Theorem 1 (the three readings of the push) are proved in full, every matrix step justified, in Appendix A. Lemma 2 (the guard $|\pi_{ST}| < \pi_T \sigma^2_{\min}$) and Proposition 2 (the saddle at $\pi_{ST} = -\pi_{TS}$) are established with the numerical diagnostics below.

Numerical diagnostics. We verify: the surprise identity $F_S(x_S^*, x_T) = \frac{\pi_{TS}}{2} x_T^\top N_S x_T$ (error $\sim 10^{-15}$); the saddle circulation $|\pi_{ST} + \pi_{TS}|\sqrt{d}$, vanishing only at $\pi_{ST} = -\pi_{TS}$; the off-manifold growth eigenvalue and terminal manifold occupancy, both turning over at the guard; the restricted novelty spectrum $\mathrm{eig}(U_T^\top N_S U_T)$ (the staircase); and, for the union and continual loops, per-bout residuals and the retention matrix. Adiabatic (fast-student) elimination is used throughout for the reduced teacher dynamics of Eq. (10). Simulations use the timescale ordering $\tau_S \ll \tau_T \ll 1/\eta$, isotropic teacher noise ξ, and a fixed teacher norm; full parameter tables accompany each experiment notebook.

Supporting information

S1. Stability and timescale sweeps. On-manifold consolidation vs. noise-chasing across $|\pi_{ST}|$ (the guard turnover); spectral, not temporal, selection across τ_S/τ_T. **S2. Effective-rank detail.** Staircase step count vs. pattern correlation. **S3. Dendritic (stop-gradient) variant.** The mechanism under Tang et al.'s bio-plausible stop-gradient relaxation [12]: transfer still completes, at the cost of a $\sim 2\times$ looser residual floor and a non-symmetric novelty operator. **S4. Noise-free slow-mixing variant.** Transfer driven by integrator imprecision rather than injected noise.

A Proofs

We give the full proofs of the four analytic results stated in the body. Every matrix step is justified explicitly. We use the house identities $M = I - W$, $S = M^\top M$, $\varepsilon_T = M_T x_T$, $\varepsilon_S = M_S x_S$, $\varepsilon_{TS} = x_S - x_T$ (Eqs. 1–2), and the free energies $F_S = \frac{\pi_{TS}}{2}\|\varepsilon_{TS}\|^2 + \frac{\pi_S}{2}\|\varepsilon_S\|^2$, $F_T = \frac{\pi_T}{2}\|\varepsilon_T\|^2$ (Eq. 6). Two algebra facts recur: **(R1)** $(AB)^\top = B^\top A^\top$; and **(R2)** for $g = \frac{1}{2}\|u\|^2$, a perturbation $u \to u + \delta$ gives $g \to g + u^\top \delta + O(\|\delta\|^2)$, so $\nabla g = u$.

Timescales. Two distinct "student fixed" assumptions are used, on the nested separation $\tau_S \ll \tau_T \ll 1/\eta$. Lemma 1 uses the student *state* being *fast*: x_S is already settled at every teacher configuration ($\dot{x_S} = 0$). Theorem 1 uses the student *weights* being *slow*: over the teacher's exploration W_S, and hence S_S and N_S, are frozen matrices. The two are not the same assumption.

A.1 Proposition 1: inference and learning descend F_S

Part A — inference. We first derive the one differentiation rule. For fixed A and $g(x) = \frac{1}{2}\|Ax\|^2 = \frac{1}{2}(Ax)^\top(Ax)$, perturb $x \to x + \delta$ and expand exactly:

$$g(x+\delta) = \tfrac{1}{2}(Ax)^\top(Ax) + (Ax)^\top(A\delta) + \tfrac{1}{2}(A\delta)^\top(A\delta).$$

The linear term is $(Ax)^\top A\delta = x^\top A^\top A\,\delta$ by (R1); its coefficient transposed gives

$$\nabla_x\, \tfrac{1}{2}\|Ax\|^2 = A^\top A\, x. \tag{12}$$

(The $\frac{1}{2}$ cancels the factor 2 from the two equal cross terms.) Apply Eq. (12) to each term of F_S, holding x_T fixed. The interface term $\frac{\pi_{TS}}{2}\|x_S - x_T\|^2$ is the form with $A = I$ and a constant shift (which drops), giving $\pi_{TS}\,\varepsilon_{TS}$. The self term $\frac{\pi_S}{2}\|M_S x_S\|^2$ is the form with $A = M_S$, giving $\pi_S M_S^\top \varepsilon_S$. Summing,

$$\nabla_{x_S} F_S = \pi_{TS}\,\varepsilon_{TS} + \pi_S\, M_S^\top \varepsilon_S,$$

and the student's state equation (4) is exactly $\tau_S \dot{x_S} = -\nabla_{x_S} F_S$: inference is exact gradient descent on F_S.

Part B — learning. Only the self term $\frac{\pi_S}{2}\|x_S - W_S x_S\|^2$ depends on W_S. Writing it as $\frac{\pi_S}{2}\sum_i \varepsilon_{S,i}^2$ with $\varepsilon_{S,i} = x_{S,i} - \sum_k (W_S)_{ik} x_{S,k}$, and using $\partial(W_S)_{ik}/\partial(W_S)_{ab} = \delta_{ia}\delta_{kb}$,

$$\frac{\partial \varepsilon_{S,i}}{\partial(W_S)_{ab}} = -\delta_{ia}\, x_{S,b} \quad \Longrightarrow \quad \frac{\partial F_S}{\partial(W_S)_{ab}} = \pi_S \sum_i \varepsilon_{S,i}(-\delta_{ia} x_{S,b}) = -\pi_S\, \varepsilon_{S,a}\, x_{S,b},$$

the δ_{ia} collapsing the sum. Assembling the entries into a matrix gives the outer product $\nabla_{W_S} F_S = -\pi_S\, \varepsilon_S\, x_S^\top$. The zero-diagonal constraint restricts admissible updates to the subspace Z of zero-diagonal matrices; the orthogonal projection onto Z is $P_0(X) = X - \operatorname{diag}(X)$, which is idempotent ($P_0^2 = P_0$) and self-adjoint ($\langle P_0 X, Y\rangle = \langle X, P_0 Y\rangle$) for the Frobenius inner product $\langle X, Y\rangle = \operatorname{tr}(X^\top Y)$. The learning rule follows the projected gradient $\dot{W}_S = -\eta\, P_0(\nabla_{W_S} F_S)$, so with $G = \nabla_{W_S} F_S$,

$$\dot{F}_S = \langle G, \dot{W}_S\rangle = -\eta\, \langle G, P_0 G\rangle = -\eta\, \langle P_0 G, P_0 G\rangle = -\eta\, \|P_0 G\|_F^2 \le 0,$$

inserting $P_0 G = P_0(P_0 G)$ then moving one P_0 across. Learning is projected gradient descent on F_S. ■

A.2 Lemma 1: the novelty operator

Step 1–2 — solve the fast-student steady state. "Fast student" means $\nabla_{x_S} F_S = 0$, i.e. $\pi_{TS}(x_S - x_T) + \pi_S S_S x_S = 0$. Collecting x_S on the left with $B := \pi_{TS} I + \pi_S S_S$,

$$B\, x_S = \pi_{TS}\, x_T \quad \Longrightarrow \quad x_S^* = \pi_{TS}\, B^{-1} x_T.$$

Step 3 — B is invertible. If $S_S v = \mu v$ then $Bv = (\pi_{TS} + \pi_S \mu)v$, so B shares the eigenbasis of S_S with eigenvalues $\pi_{TS} + \pi_S\mu \ge \pi_{TS} > 0$ (using $\mu \ge 0$ from Step 5). None is zero, so B^{-1} exists and $B^{-1}v = (\pi_{TS} + \pi_S\mu)^{-1} v$.

Step 4 — the leftover interface error. Writing $x_T = B^{-1} B x_T$ to share a left factor,

$$\varepsilon_{TS} = x_S^* - x_T = B^{-1}\big[\pi_{TS} I - (\pi_{TS} I + \pi_S S_S)\big] x_T = B^{-1}(-\pi_S S_S) x_T = -\pi_S\, B^{-1} S_S\, x_T.$$

Step 5 — S_S is symmetric PSD, and learned ⇒ zero. For any v, $v^\top S_S v = v^\top M_S^\top M_S v = \|M_S v\|^2 \ge 0$, so S_S is PSD ($\mu \ge 0$) and symmetric ($(M_S^\top M_S)^\top = M_S^\top M_S$). A learned direction u has $M_S u = 0$, hence $S_S u = M_S^\top M_S u = 0$.

Step 6–7 — S_S commutes with B^{-1}; the operator N_S. Since $S_S B = \pi_{TS} S_S + \pi_S S_S^2 = B S_S$, sandwiching by B^{-1} gives $B^{-1} S_S = S_S B^{-1}$. Sliding S_S left in Step 4,

$$\varepsilon_{TS} = -\pi_S\, S_S\, B^{-1} x_T = -N_S\, x_T, \qquad N_S = \pi_S\, S_S\, (\pi_{TS} I + \pi_S S_S)^{-1}.$$

N_S is symmetric (product of the commuting symmetric S_S and B^{-1}: $(S_S B^{-1})^\top = B^{-1} S_S = S_S B^{-1}$) and PSD (eigenvalues $n(\mu) = \pi_S \mu/(\pi_{TS} + \pi_S \mu) \ge 0$).

Step 8 — N_S annihilates learned directions. For learned u, Step 5 gives $S_S u = 0$, so $N_S u = \pi_S B^{-1} S_S u = 0$ and $\varepsilon_{TS} = -N_S u = 0$: a teacher along a learned direction produces no drive. ■

Complement identity. From $BB^{-1}=I$, i.e. $\pi_{TS}B^{-1}+\pi_S S_S B^{-1}=I$, and the definition of N_S, we get $\pi_{TS}B^{-1}=I-N_S$, hence

$$x_S^*=(I-N_S)\,x_T,\qquad \varepsilon_{TS}=(I-N_S)x_T-x_T=-N_S x_T.$$

The student copies the predicted part (I) and leaves the novel part ($-N_S$): per direction it keeps the fraction $1-n_k$ and leaves n_k behind as error.

A.3 Corollary 1: the surprise identity

Let the student settle ($x_S=x_S^*$) with the teacher at x_T, and read off $F_S^{\mathrm{eq}}(x_T):=F_S(x_S^*(x_T),x_T)$; since F_S is a strictly convex quadratic in x_S (Hessian $B\succ 0$), this equals $\min_{x_S}F_S$.

Direct computation. Plug in $x_S^*=(I-N_S)x_T$ and $\varepsilon_{TS}=-N_S x_T$. The interface term, using N_S symmetric,

$$\tfrac{\pi_{TS}}{2}\|\varepsilon_{TS}\|^2=\tfrac{\pi_{TS}}{2}\,x_T^\top N_S^2\,x_T.$$

The self term, using $\|\varepsilon_S\|^2=(x_S^*)^\top S_S\,x_S^*$ and the key move $\pi_S S_S(I-N_S)=\pi_{TS}(\pi_S S_S B^{-1})=\pi_{TS}N_S$ (complement identity),

$$\tfrac{\pi_S}{2}\|\varepsilon_S\|^2=\tfrac{\pi_S}{2}\,x_T^\top(I-N_S)S_S(I-N_S)x_T=\tfrac{\pi_{TS}}{2}\,x_T^\top(N_S-N_S^2)x_T.$$

Adding, the N_S^2 terms cancel:

$$F_S^{\mathrm{eq}}(x_T)=\tfrac{\pi_{TS}}{2}\,x_T^\top\big[N_S^2+N_S-N_S^2\big]x_T=\tfrac{\pi_{TS}}{2}\,x_T^\top N_S\,x_T.$$

Per direction $x_T=\sum_k c_k u_k$: interface costs n_k^2, self costs $n_k(1-n_k)$, summing to n_k; so $F_S^{\mathrm{eq}}=\frac{\pi_{TS}}{2}\sum_k n_k c_k^2$ — novelty priced on where the state sits. ■

Gradient (envelope route). For $g(\theta)=\min_x f(x,\theta)=f(x^*(\theta),\theta)$ the chain rule gives $g'(\theta)=\underbrace{(\partial f/\partial x)|_{x^*}}_{=0}\,dx^*/d\theta+(\partial f/\partial\theta)|_{x^*}$, the first factor vanishing by the definition of the minimizer. Here $\theta=x_T$ and only the interface term carries x_T explicitly, with x_T-gradient $\pi_{TS}(x_T-x_S)$; freezing $x_S=x_S^*$,

$$\nabla_{x_T}F_S^{\mathrm{eq}}=\pi_{TS}(x_T-x_S^*)=-\pi_{TS}\,\varepsilon_{TS}=\pi_{TS}\,N_S\,x_T,$$

matching the direct computation.

A.4 Theorem 1: prioritized transfer

Step 1 — read off the push. The teacher in sleep obeys $\tau_T\dot{x_T}=-\pi_T M_T^\top\varepsilon_T+\pi_{ST}\varepsilon_{TS}+\xi$. Substituting $M_T^\top\varepsilon_T=S_T x_T$, the fast-student $\varepsilon_{TS}=-N_S x_T$ (Lemma 1), dropping ξ, and writing $\pi_{ST}=-|\pi_{ST}|$,

$$\tau_T\dot{x_T}=\underbrace{-\pi_T S_T x_T}_{\text{self-pull}}+\underbrace{|\pi_{ST}|N_S x_T}_{u},\qquad u=|\pi_{ST}|\,N_S\,x_T.$$

Step 2 — per-direction amplification. N_S is symmetric, so $N_S u_k=n_k u_k$ on an orthonormal eigenbasis, $n_k\in[0,1)$, $n_k=0$ iff learned. With $c_k=u_k^\top x_T$ and under the push alone ($\dot{x_T}=u/\tau_T$),

$$\dot c_k=u_k^\top\dot{x_T}=\frac{|\pi_{ST}|}{\tau_T}\,u_k^\top N_S x_T=\frac{|\pi_{ST}|}{\tau_T}\,n_k\,c_k,$$

using $u_k^\top N_S=n_k u_k^\top$. Learned directions ($n_k=0$) are frozen; novel ones grow, faster the more novel.

Step 3 — steepest ascent of the student's surprise. By Corollary 1, $F_S^{\mathrm{eq}}(x_T) = \frac{\pi_{TS}}{2} x_T^\top N_S x_T = \frac{\pi_{TS}}{2} \sum_k n_k c_k^2$, whose gradient (by perturbation, as in Eq. 12, N_S symmetric) is $\nabla_{x_T} F_S^{\mathrm{eq}} = \pi_{TS} N_S x_T$. Hence

$$u = |\pi_{ST}|\, N_S\, x_T = \frac{|\pi_{ST}|}{\pi_{TS}}\, \nabla_{x_T} F_S^{\mathrm{eq}},$$

a positive multiple of the surprise gradient. By Cauchy–Schwarz $\nabla F_S^{\mathrm{eq}} \cdot e \le \|\nabla F_S^{\mathrm{eq}}\|$ with equality iff $e \parallel \nabla F_S^{\mathrm{eq}}$: of all directions, the push raises the student's surprise fastest (a local steepest ascent).

Step 4 — climbing and self-limiting. Along the push,

$$\frac{dF_S^{\mathrm{eq}}}{dt} = \nabla F_S^{\mathrm{eq}} \cdot \dot{x_T} = \frac{\pi_{TS}|\pi_{ST}|}{\tau_T}\, \|N_S x_T\|^2 \ge 0,$$

zero only when $N_S x_T = 0$. And $n(\mu) \le (\pi_S/\pi_{TS})\mu$ gives $|\pi_{ST}| n_k \le (|\pi_{ST}|\pi_S/\pi_{TS})\|M_S u_k\|^2$, so as the student learns a direction ($\|M_S u_k\| \to 0$, Proposition 1) the push there vanishes: the steering self-terminates direction by direction.

Step 5 — the saddle condition (preview of Proposition 2). The self-pull is $-\nabla_{x_T} F_T$ with $\nabla_{x_T} F_T = \pi_T S_T x_T$, and the push is $(|\pi_{ST}|/\pi_{TS})\nabla_{x_T} F_S^{\mathrm{eq}}$, so

$$\tau_T \dot{x_T} = -\nabla_{x_T} F_T + \frac{|\pi_{ST}|}{\pi_{TS}}\, \nabla_{x_T} F_S^{\mathrm{eq}},$$

which is $-\nabla\Phi$ for the single potential $\Phi = F_T - F_S^{\mathrm{eq}}$ exactly when the ascent coefficient is one: $|\pi_{ST}| = \pi_{TS}$, i.e. $\pi_{ST} = -\pi_{TS}$. The ascent is local, and the push rescales but does not seed an unoccupied direction ($c_k = 0 \Rightarrow \dot{c}_k = 0$); noise supplies that seed in the full model. ∎

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